# Aaron Lin

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## EDUCATION

**Boston University** Boston, MA
*B.S. Mechanical Engineering, Aerospace Concentration, Materials Science & Engineering Minor* Sep 2024 – May 2028
GPA: 3.76/4.0, Dean's List
**Completed Coursework:** Mechanics of Materials, Materials Science, Energy and Thermodynamics, CAD and Machine Components, Probability and Statistics, Differential Equations, Computational Linear Algebra
**Current (Fall 2026):** Fluid Mechanics, Dynamics, Electromechanical Design, Manufacturing Processes, Thermodynamics and Statistical Mechanics (Material Science Elective)
**Planned (Spring 2027):** Aerodynamics, Measurement and Instrumentation, Structural Mechanics, Heat Transfer

## PROFESSIONAL EXPERIENCE

**Mechanical Engineering Intern** Feb 2026 – Jun 2026
*Culturon (plasma-activated coatings for life sciences)* Sydney, Australia
- Designed and executed Optical Emission Spectroscopy (OES) experiments using an HR4000 spectrometer; authored repeatable protocols for plasma consistency evaluation and elemental contaminant detection.
- Developed an automated Python analysis suite (pandas, SciPy, Astroquery) to process OES spectral data, isolate elemental emission peaks, and cross-reference identifications against the NIST atomic line database.
- Authored a benchmarking report comparing OES plasma diagnostics against an established reference chamber and documenting key emission-line signatures.

**AI Trainer** Oct 2025 – Dec 2025
*Handshake AI Fellowship* Remote
- Developed and evaluated domain-specific prompts benchmarking LLM accuracy across engineering and physical science subfields; applied subject-matter expertise to identify reasoning failures on scientific and quantitative topics.

## PROJECTS

**Project Engineer** Sep 2024 – Present
*Boston University Rocket Propulsion Group* Boston, MA
- Designing an aerodynamic fairing for an externally-mounted downcomer on the ARGO rocket; evaluating tangent-ogive and power-law nosecone profiles at a Mach 2.81 design condition using ANSYS Fluent CFD and OpenRocket trajectory simulation.
- Designed a converging-diverging rocket nozzle in SolidWorks using 1D compressible-flow theory; created part and assembly drawings and performed pressure-load FEA for an 80 psi chamber condition across three design reviews.
- Fabricated the nozzle via resin 3D printing and helped integrate it with a 1-inch NPT chamber, seals, relay ignition circuitry, and data acquisition for a team static fire.

**CAD and Mechanical Layout Lead** Sep 2026 – Present
*2.5-DOF Cartesian Fishing Game* Boston, MA
- Developing the CAD and mechanical layout for a joystick-controlled 2.5-DOF mechanism, translating carriage, travel, packaging, and assembly requirements into the final design.
- Team-built lead-screw prototypes demonstrated coordinated X-Y motion and Z-axis actuation in square and circle path tests; final assembly and testing is ongoing.

**Firmware and Integration Contributor** Sep 2025 – Dec 2025
*Autonomous Desk Cleaning Robot* Boston, MA
- Authored Arduino C++ firmware implementing S-curve pathing, IR proximity-triggered 90° edge-avoidance turns, and gesture-activated start to enable autonomous operation across a 4 ft by 4 ft desk surface.
- Team ranked in the top 4 of approximately 20 groups; the documented final trial collected 72 of 95 rice grains with the upgraded fan and nozzle configuration.

## SKILLS

**Software:** SolidWorks (Design, Simulation, HSMWorks), Fusion, Onshape, ANSYS (Fluent, Mechanical, Workbench), COMSOL, OpenRocket, Python, MATLAB, LaTeX, C++, Java
**Engineering:** Mechanical Design, FEA, CFD, CAD, OES, Supersonic Aerodynamics, 1D Compressible Flow, Stress Analysis, GD&T (fundamentals), Thermodynamic Cycle Analysis, Technical Writing, Systems Integration, Experimental Testing, Arduino
**Fabrication:** Additive Manufacturing (FDM, Resin), CNC Machining,Manual Lathe, Manual Mill, Soldering, Electronics Assembly
**Languages:** English (Native), Mandarin Chinese (Conversational)

## CERTIFICATIONS

A Hands-On Introduction to Engineering Simulations — edX / CornellX, 2026
Machine Learning Specialization — DeepLearning.AI / Stanford University, 2024